Kuan-Chia, Chen

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Education

Oregon State University — Corvallis, OR
M.S. in Computer Science — *Sep 2023 – Jun 2026*

National Taipei University of Technology — Taipei, Taiwan
B.S. in Electrical Engineering — *Sep 2019 – Jun 2023*

Experience

Dynamic Robotics and Artificial Intelligence Laboratory — Corvallis, OR
Graduate Research Assistant (Advisor: Prof. Alan Fern) — *Oct 2023 – Jun 2026*

- Proposed and implemented a keypoint-based reinforcement learning (RL) policy for retargeting human motion to the Digit V3 humanoid robot, achieving over 13 minutes of continuous VR teleoperation.
- Explored the application of diffusion policies for humanoid manipulation using VR-collected demonstrations.
- Developed a hierarchical RL policy for box pick-up using structured object representations (pose and size), improving real-world success rate from 60% to 90% while reducing training time by 50%.
- Designed a hierarchical RL policy for box put down, achieving 93% success in simulation and 90% on hardware.
- Developed a point cloud-based pick-up policy for robust box pickup across varying poses in Isaac Lab.

Puti Intelligence — Corvallis, OR
Research Scientist Intern (Loco-Manipulation) — *Aug 2025 – Sep 2025*

- Developed hierarchical manipulation policies and improved the teleoperation system, increasing stable operation time from 20 minutes to over 1 hour for real-world product demonstrations.
- Deployed RGB-D cameras on a humanoid robot, mitigating sim-to-real gaps caused by depth noise.
- Built an RGB-D pipeline for 9-DoF object pose estimation and integrating it with RL-based manipulation policies.

Innoveon — Taipei, Taiwan
Robotics Engineer Intern (Autonomous Systems) — *Feb 2023 – Jun 2023*

- Developed autonomous mobile robots (AMR) using ROS for navigation, planning, and control.
- Implemented SLAM algorithms for real-time localization and mapping in dynamic environments.

Publications

Humanoid Hanoi: Shared Whole-Body Control for Long-Horizon Box Rearrangement — *2026*
Under review at Robotics: Science and Systems (RSS 2026) | Project Page: link
Minku Kim†, **Kuan-Chia Chen**†, Aayam Shrestha, Li Fuxin, Stefan Lee, Alan Fern

Research Projects

Point Cloud to Action: Hierarchical Manipulation Policy — Corvallis, OR

- Trained hierarchical RL policies on the Digit V3 humanoid robot in Isaac Lab for box pick-up tasks. — *2025*
- Learned manipulation policies directly from point cloud inputs (PointNet), without explicit object pose estimation.

Diffusion Policy for Robot Manipulation — Corvallis, OR

- Trained diffusion-based policies on VR-collected real-world demonstrations for data-efficient robot learning. — *2025*
- Evaluated action rollout horizons (1–8 steps) for performance-speed tradeoffs in MuJoCo.

Keypoint Mimic Policy for VR-Based Whole Body Control — Corvallis, OR

- Designed trajectory-based rewards for an RL policy (PPO + LSTM) for humanoid motion retargeting. — *2024*
- Built a VR teleoperation system and mapped human keypoints and locomotion signals to policy inputs.
- Achieved real-time (50Hz) control with over 13 minutes of continuous task execution on humanoid hardware.

Skills

Robotics & AI: Bipedal Locomotion, Manipulation, Whole-Body Control, Sim-to-Real Transfer, Reinforcement Learning, Imitation Learning, Diffusion, Motion Retargeting, SLAM
Programming and simulation: MuJoCo, Isaac Lab, Python, Pytorch, C++, CUDA, Arduino
Hardware and Systems: Jetson Xavier NX, Embedded Systems, PLC, ROS, LIDAR

Awards

Merit-Based International Graduate Scholarship, Oregon State University — *2023*
First Place, Taiwan National Senior Secondary School Industrial Control Competition — *2018*